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Hardware

Nano 33 BLE

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This tutorial refers to a product that has reached its end-of-life status.

Accessing Accelerometer Data on Nano 33 BLE

Learn how to measure the relative position of the Nano 33 BLE through the LSM9DS1 IMU sensor.

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Alushi · revision

This tutorial will focus on the 3-axis accelerometer sensor of the **LSM9DS1** module on the Arduino Nano 33 BLE, in order to measure the relative position of the board. This will be achieved by utilizing the values of the accelerometer's axes and later print the return values through the Arduino IDE Serial Monitor.

Goals

The goals of this project are:

Understand what an LSM9DS1 module is.

Use the LSM9DS1 library.

Read the raw data of the

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Goals

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The LSM9DS1 Inertial Module

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Convert the raw data into board positions.

Print out live data through the Serial Monitor.

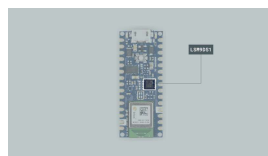
Hardware & Software Needed

This project uses no external sensors or components.

In this tutorial we will use the Arduino Create Cloud Editor to program the board.

The LSM9DS1 Inertial Module

IMU stands for: inertial measurement unit. It is an electronic device that measures and reports a body's specific force, angular rate and the orientation of the body, using a combination of accelerometers, gyroscopes, and oftentimes magnetometers. In this tutorial we will learn about the LSM9DS1 IMU module, which is included in the Arduino Nano 33 BLE Board.



The LSM9DS1 sensor.

The LSM9DS1 is a system-in-package featuring a 3D digital

digital angular rate sensor, and a 3D digital magnetic sensor.

The LSM9DS1 Library

The Arduino LSM9DS1 library allows us to use the Arduino Nano 33 BLE IMU module without having to go into complicated programming. The library takes care of the sensor initialization and sets its values as follows:

Accelerometer range is set at $[-4, +4]g$ ± 0.122 mg.

Gyroscope range is set at $[-2000, +2000]$ dps ± 70 mdps.

Magnetometer range is set at $[-400, +400]$ uT ± 0.014 uT.

Accelerometer output data rate is fixed at 104 Hz.

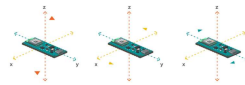
Gyroscope output data rate is fixed at 104 Hz.

Magnetometer output data rate is fixed at 20 Hz.

If you want to read more about the LSM9DS1 sensor module see [here](#).

Accelerometer

An accelerometer is an electromechanical device used to measure acceleration forces. Such forces may be static, like the continuous force of gravity or, as is the case with many mobile devices, dynamic to sense movement or vibrations.



How the
accelerometer works.

In this example, we will use the accelerometer as a "level" that will provide information about the position of the board. With this application we will be able to read what the relative position of the board is as well as the degrees, by tilting the board up, down, left or right.

Creating the Program

1. Setting up

Let's start by opening the Arduino Cloud Editor, click on the **Libraries** tab and search for the **LSM9DS1** library. Then in **Examples**, open the **SimpleAccelerometer** sketch and once it opens, rename it as **Accelerometer**.



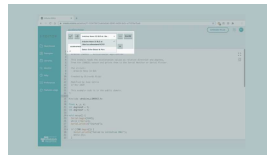
Finding the library in
the Cloud Editor.

2. Connecting the board

Now, connect the Arduino Nano
33 BLE to the computer and

recognizes it, if so, the board and port should appear as shown in the image below. If they don't appear, follow the [instructions](#) to install the plugin that will allow the Editor to recognize your board.

Now, connect the Arduino Nano 33 BLE to the computer and make sure that the Cloud Editor recognizes it. If so, the board and port should appear as shown in the image below. If they don't appear, follow the [instructions](#) to install the plugin that will allow the Editor to recognize your board.



Selecting the board.

3. Writing the Code

Now we will write the code to read the accelerometer data, calculate the tilt angles, and print the relative position of the board as we move it at different angles.

Include the LSM9DS1 library at the top of your sketch:

```
1 #include <Arduino_LSM9DS1>
```

Initialize variables before the

```
1 #define MINIMUM_TILT 5
2 #define WAIT_TIME 500
3
4 float x, y, z;
5 int angleX = 0;
6 int angleY = 0;
7 unsigned long previous
```

In the `setup()`, initialize the IMU and start serial communication:

```
1 void setup() {
2   Serial.begin(9600);
3   while (!Serial);
4
5   if (!IMU.begin()) {
6     Serial.println("F
7     while (1);
8   }
9
10  Serial.print("Accel
11  Serial.print(IMU.ac
12  Serial.println(" Hz
13 }
```

Write the `loop()` function to read accelerometer data and calculate tilt angles:

```

1 void loop() {
2   unsigned long curTime = millis();
3
4   if (IMU.accelerati
5     previousMillis = curTime;
6
7   IMU.readAccelerat
8
9   // Calculate tilt
10  angleX = atan2(>
11  angleY = atan2(>
12
13  // Determine the
14  if (angleX > MIN
15    Serial.print('
16    Serial.print(
17    Serial.println
18  } else if (angle
19    Serial.print('
20    Serial.print(
21    Serial.println
22  }
23
24  if (angleY > MIN
25    Serial.print('
26    Serial.print(
27    Serial.println
28  } else if (angle

```

In this code, we use trigonometric functions to calculate the tilt angles from the accelerometer data. The `MINIMUM_TILT` variable is used to ignore small movements below a certain threshold.

Note: For the following code to work properly, the board's facing direction and inclination during the initialization of the code need to be specific. More information will be shared in the "Testing It Out" section.

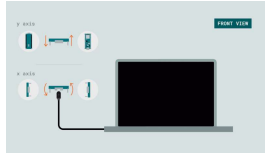
Complete Code

If you choose to skip the code-building section, the complete code can be found below:

```
1  /*
2   Arduino LSM9DS1 -
3
4   This example reads
5   from the LSM9DS1 s
6
7   The circuit:
8   - Arduino Nano 33
9
10  Created by Pedro L
11
12  This example code
13  */
14
15  #include <Arduino_L
16
17  #define MINIMUM_TIL
18  #define WAIT_TIME 50
19
20  float x, y, z;
21  int angleX = 0;
22  int angleY = 0;
23  unsigned long previo
24
25  void setup() {
26    Serial.begin(9600);
27    while (!Serial);
28
```

Testing It Out

In order to get a correct reading of the board data, before uploading the sketch to the board hold the board in your hand, from the side of the USB port. The board should be facing up and "pointing" away from you. The image below illustrates the board's position and how it

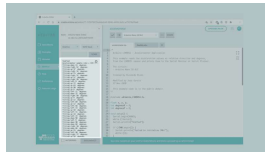


Interacting with the X
and Y axes.

Now, you can verify and upload the sketch to the board and open the Monitor from the menu on the left.

If you tilt the board upwards, downwards, right or left, you will see the results printing every second according to the direction of your movement!

Here is a screenshot of the sketch returning these values:



Printing out the "tilt
condition" of the
board.

Troubleshoot

Sometimes errors occur, if the code is not working there are some common issues we can troubleshoot:

- Missing a bracket or a semicolon.

- Arduino board connected to the wrong port.

- Accidental interruption of cable connection.

The initial position of the board is not as instructed. In this case you can refresh the page and try again.

Conclusion

In this simple tutorial we learned what an IMU sensor module is, how to use the **LSM9DS1** library, and how to use an Arduino Nano 33 BLE to get data. Furthermore, we utilized the 3-axis accelerometer sensor, in order to measure and print out the degrees and relative position of the board.

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